Rossmann Store Sales - Demand Forecasting

2026-05-08

Contents

Executive Summary

This project builds a demand forecasting model using the Rossmann Store Sales dataset. Four models were developed and compared progressively, from a naive store-average baseline through linear regression, decision trees, and random forests. The best-performing model, linear regression with store-level fixed effects, achieved an RMSE of approximately 1,370, reducing prediction error by 28% over the baseline.

1 - Introduction

"Retail is Detail!"

Retail demand forecasting is one of the most practically valuable applications of machine learning in supply chain management. Knowing how much stock a store will need, days, weeks, or months in advance, directly impacts inventory levels, staffing decisions, and supplier orders.

This project uses the Rossmann Store Sales dataset, sourced from a Kaggle competition hosted by Rossmann, one of Europe's largest drug store chains. The dataset contains 1,017,209 daily sales records across 1,115 stores spanning from January 2013 to July 2015. Each record includes information about promotions, holidays, and store-level characteristics such as store type, assortment level, and competitor proximity.

The objective is to predict daily sales (`Sales`) for each store-date combination. Model performance is evaluated using Root Mean Square Error (RMSE):

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}$$

A lower RMSE indicates better predictive accuracy.

The report is structured as follows: Section 2 describes the analytical approach, Section 3 presents exploratory data analysis, Section 4 covers feature engineering, Section 5 details the modelling process, Section 6 presents results, and Section 7 concludes with limitations and future directions.

Table 1: Dataset Overview

Metric	Value
Total Rows	1,017,209
Total Columns	18
Date Range	2013-01-01 to 2015-07-31

2 - Data Familiarisation

Before any analysis, it is important to understand what each column represents and what condition the data is in.

2.1 - Column Descriptions

The merged dataset contains 18 columns drawn from two source files.

Column	Type	Description
Store	Integer	Unique store identifier (1–1115)
DayOfWeek	Integer	Day of week (1=Monday, 7=Sunday)
Date	Date	Calendar date of the record
Sales	Integer	Total sales for that day

Column	Type	Description
Customers	Integer	Number of customers that day
Open	Binary	1 = store open, 0 = store closed
Promo	Binary	1 = promotion running that day
StateHoliday	Character	a=public, b=Easter, c=Christmas, 0=none
SchoolHoliday	Binary	1 = schools closed that day
StoreType	Character	Store format (a, b, c, d)
Assortment	Character	Product range: a=basic, b=extra, c=extended
CompetitionDistance	Integer	Metres to nearest competitor store
CompetitionOpenSinceMonth	Integer	Month competitor opened
CompetitionOpenSinceYear	Integer	Year competitor opened
Promo2	Binary	1 = store runs a recurring promotion
Promo2SinceWeek	Integer	Week store joined Promo2
Promo2SinceYear	Integer	Year store joined Promo2
PromoInterval	Character	Months Promo2 is active e.g. "Jan,Apr,Jul,Oct"

2.2 - First Look at the Data

Table 2: First Six Rows of the Dataset (Key Columns)

Store	DayOfWeek	Date	Sales	Customers	Open	Promo	StoreType	Assortment	CompetitionDistance
1	5	2015-07-31	5263	555	1	1	c	a	1270
2	5	2015-07-31	6064	625	1	1	a	a	570
3	5	2015-07-31	8314	821	1	1	a	a	14130
4	5	2015-07-31	13995	1498	1	1	c	c	620
5	5	2015-07-31	4822	559	1	1	a	a	29910
6	5	2015-07-31	5651	589	1	1	a	a	310

2.3 - Missing Values

Table 3: Columns With Missing Values

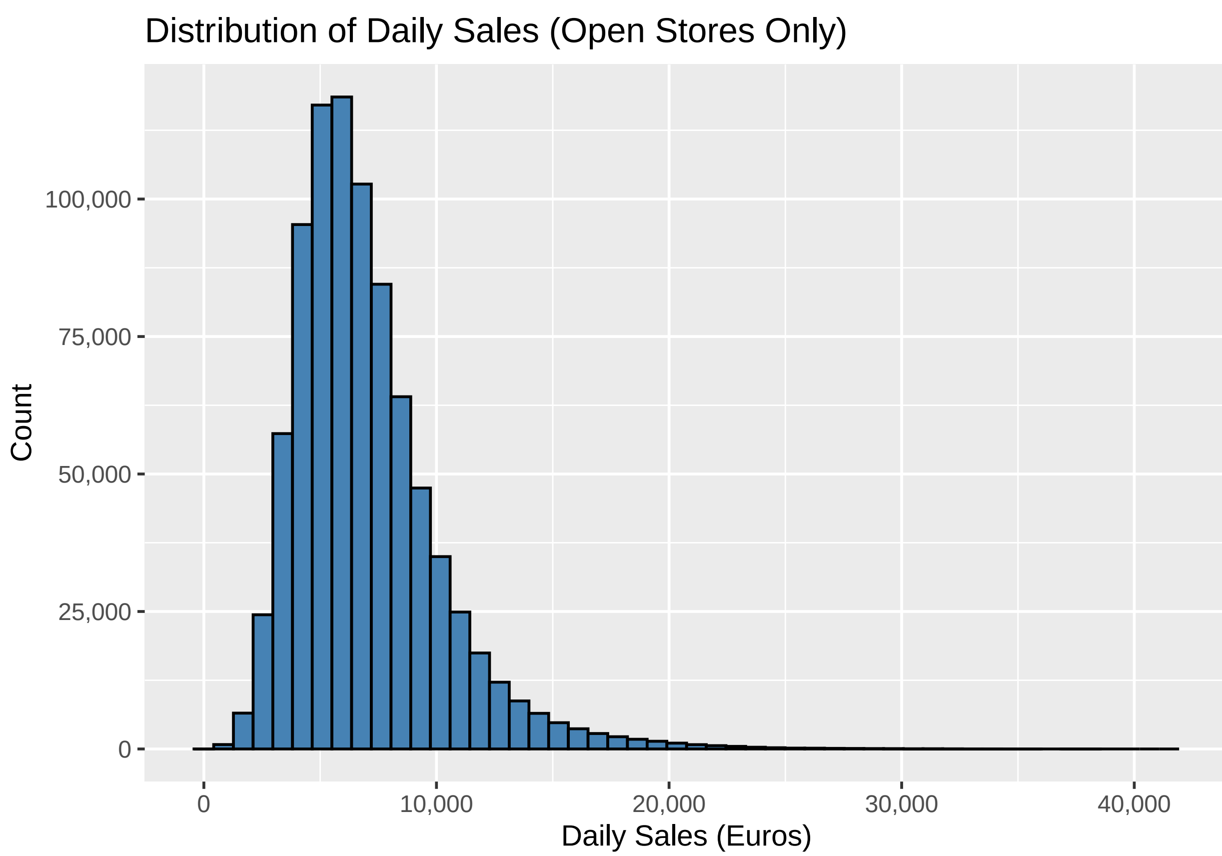
Column	Missing_Values
Promo2SinceWeek	508031
Promo2SinceYear	508031
CompetitionOpenSinceMonth	323348
CompetitionOpenSinceYear	323348
CompetitionDistance	2642

The missing values above are all in store metadata columns, not in the daily sales records themselves. This means every sales observation is intact. The missing metadata relates to competition and promotion history, which we will handle carefully in the feature engineering section.

3 - Exploratory Data Analysis

Before modelling, the training data was examined to uncover patterns in sales across time, promotions, store types, and competitor proximity. Closed store days are excluded from all analysis, a store recording zero sales because it is shut is not a demand signal.

3.1 - Sales Distribution

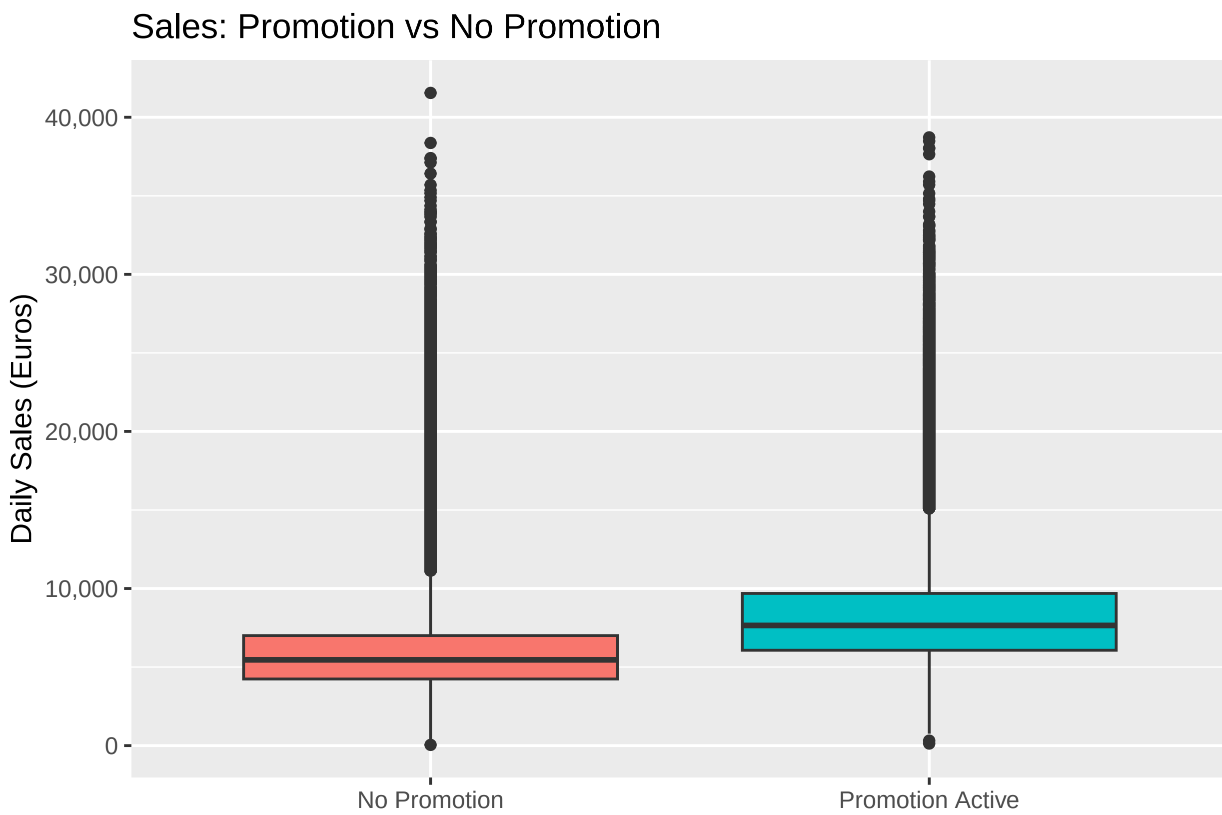


Daily sales follow a right-skewed distribution, with most store-days generating between 3,000 and 10,000 euros. A small number of exceptionally high-performing days pull the tail rightward. This skew is typical in retail sales data and suggests that a small number of stores or calendar events drive disproportionate revenue, a pattern familiar in demand planning.

3.2 - Effect of Promotions

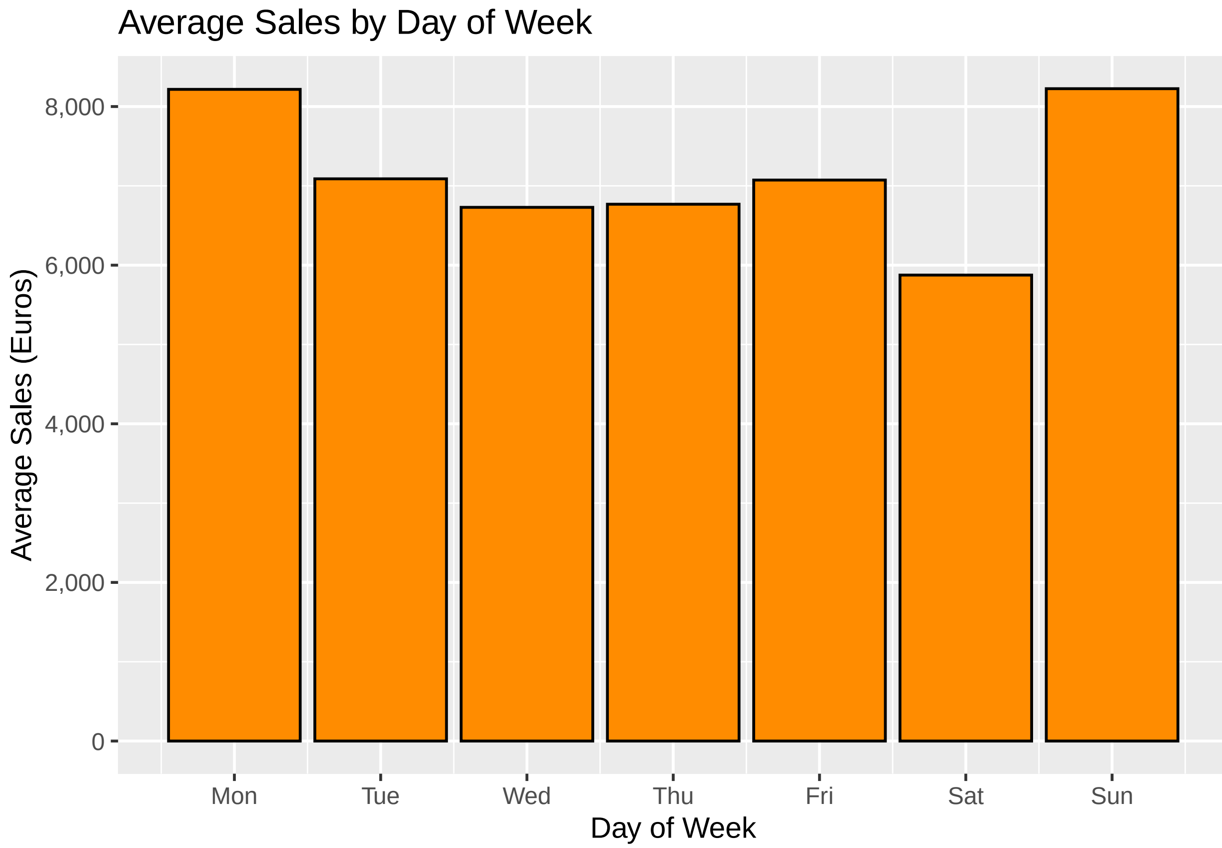
Table 4: Sales With vs Without Promotion

Promo	Avg_Sales	Median_Sales	Count
No Promotion	5930	5459	467463
Promotion Active	8229	7650	376875



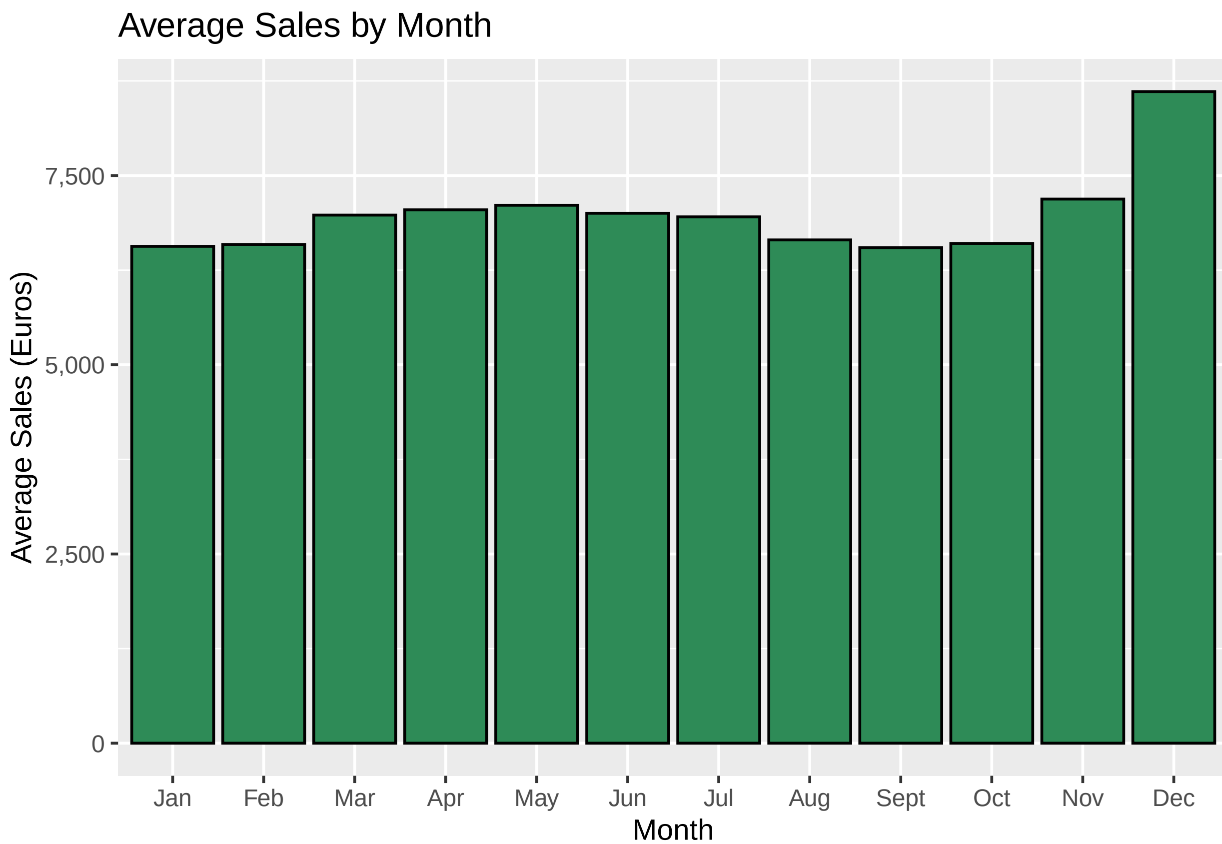
Promotions have a clear and consistent positive effect on sales. This is the strongest single signal in the dataset. From a supply chain perspective, this reinforces the importance of promotion calendars in demand planning, as promotional uplifts must be anticipated well in advance to avoid stockouts during campaign periods.

3.3 - Day of Week Patterns



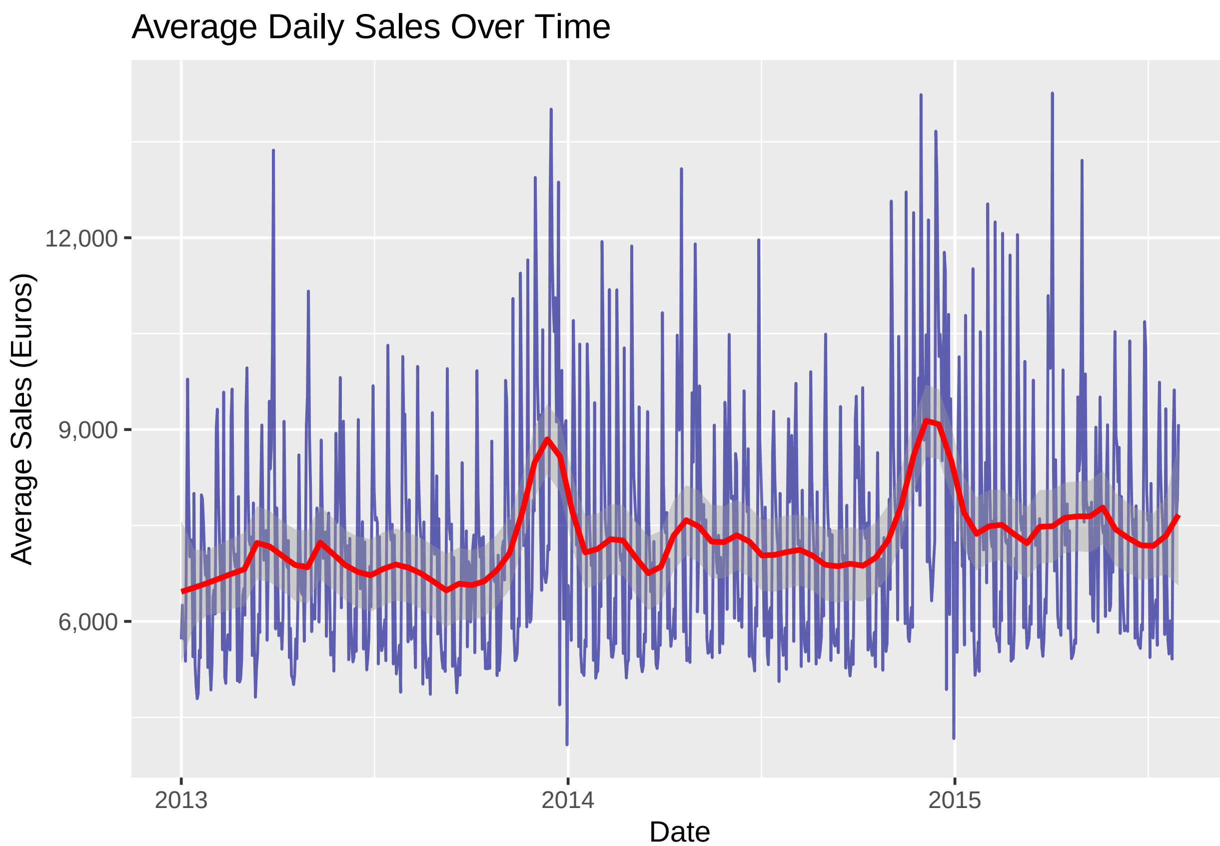
Sales peak on Mondays and Fridays and drop on Sundays, reflecting German retail trading laws that restrict Sunday opening hours. This strong day-of-week pattern is a fundamental feature in retail demand forecasting. Any model that ignores which day of the week it is will perform poorly.

3.4 - Monthly Seasonality



December shows the highest average sales, consistent with Christmas shopping behaviour. A secondary peak is visible in spring. These seasonal patterns are well understood in retail demand planning, they confirm that month of year is a necessary feature in any forecasting model.

3.5 - Sales Trend Over Time

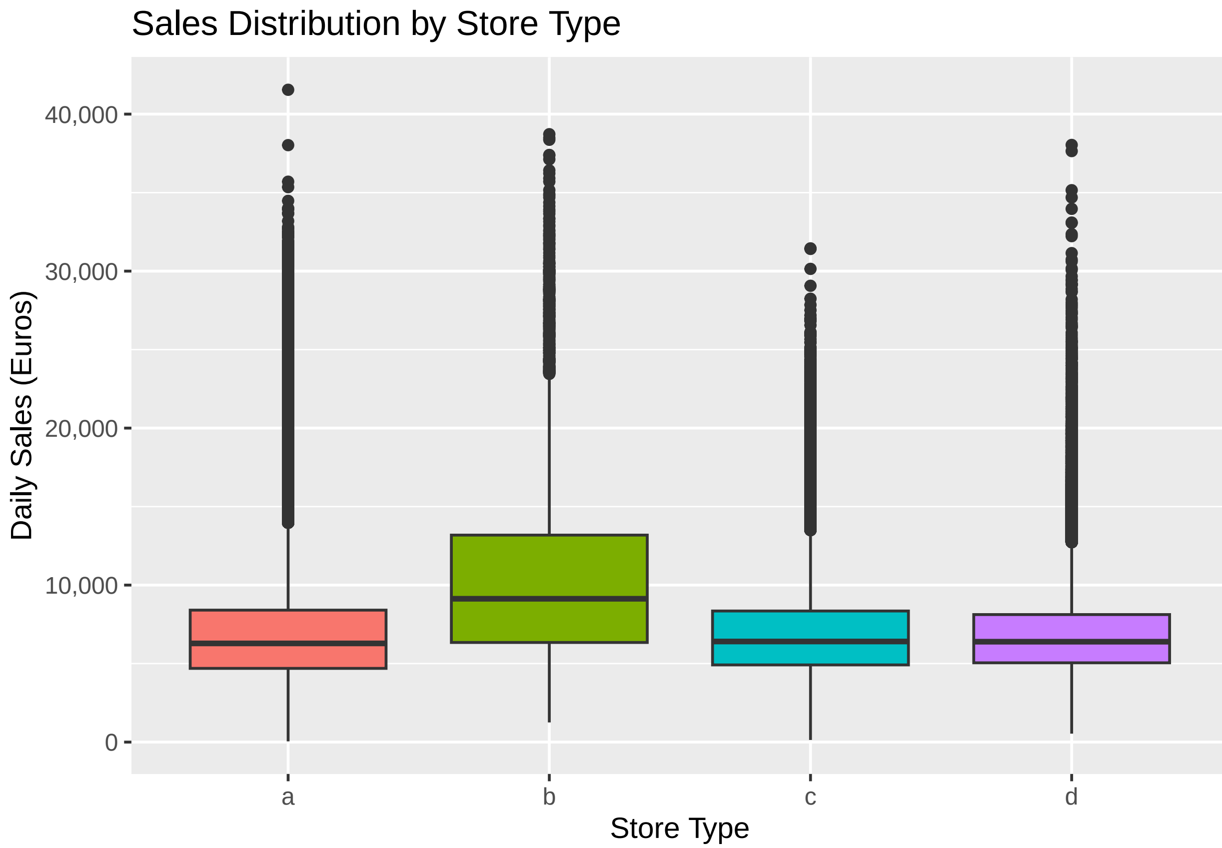


The overall sales trend is slightly upward across the two and a half year period, with strong recurring seasonal spikes each December. Short-term volatility reflects weekly cycles and promotional activity. The smooth trend line confirms steady underlying growth across the observation window.

3.6 - Store Type Comparison

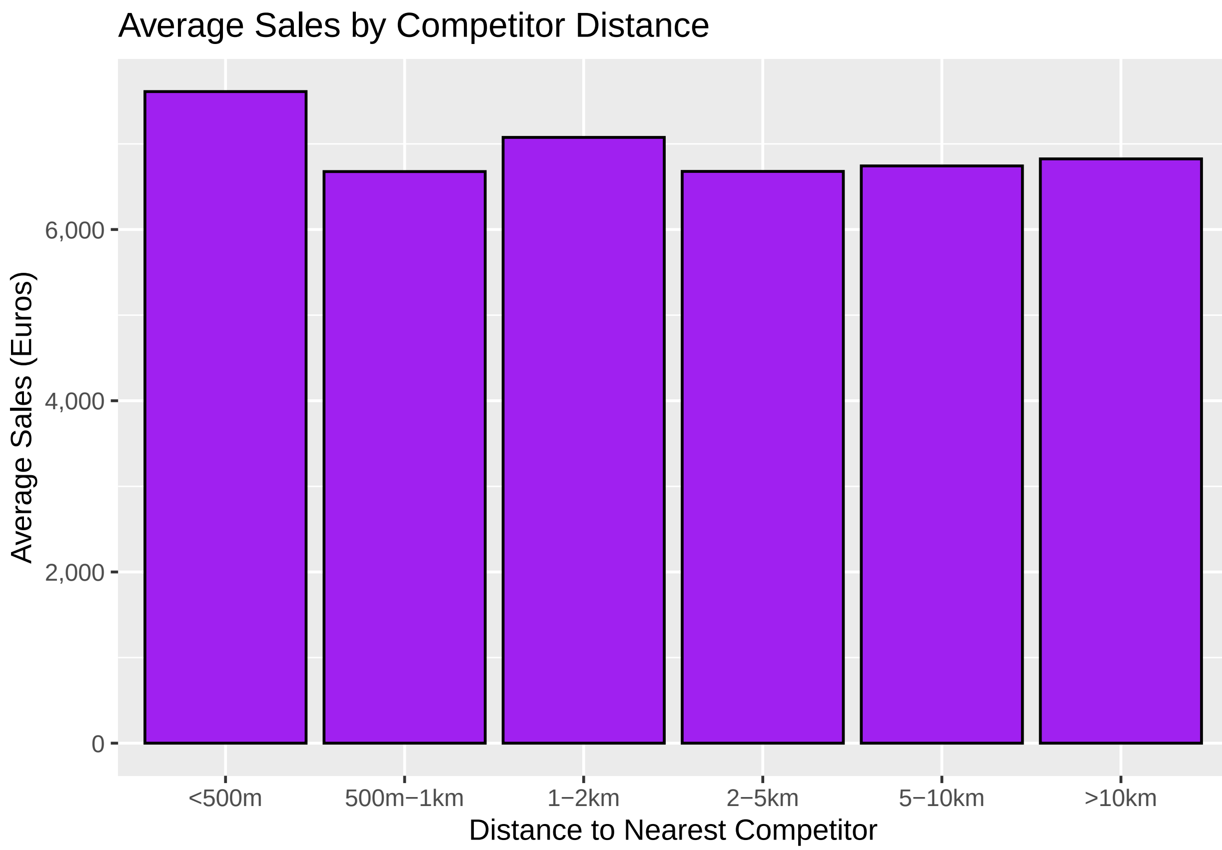
Table 5: Average Sales by Store Type

StoreType	Avg_Sales	Count
a	6926	457042
b	10233	15560
c	6933	112968
d	6822	258768



Store type B stands out with significantly higher average sales than types A, C, and D. This suggests store format is a strong predictor of baseline sales volume, likely reflecting differences in store size, location type, and product range. Store type will be included as a categorical feature in all models.

3.7 - Competitor Distance Effect



Stores with closer competitors tend to generate higher sales rather than lower. This counterintuitive finding likely reflects a location effect: stores in dense urban areas attract more competitors but also more customers. Competitor proximity is therefore a proxy for footfall rather than a direct negative pressure on sales.

3.8 - Competitor Distance: Controlling for Customers

The previous chart showed higher sales near competitors, but this could simply reflect busier locations. To separate the location effect from the competition effect, we control for customer numbers by examining sales per customer and average customer count across competitor distance bands.

If avg_customers drops near competitors but sales_per_customer stays flat, the location effect explains everything. If sales_per_customer also drops near competitors, there is a genuine competitive pricing or range pressure effect.

Table 6: Sales Per Customer by Competitor Distance

dist_bucket	avg_sales_per_customer	avg_customers	avg_sales
<500m	8.05	969	7611
500m-1km	8.73	835	6677
1-2km	9.24	798	7076

dist_bucket	avg_sales_per_customer	avg_customers	avg_sales
2-5km	9.72	692	6679
5-10km	10.85	630	6743
>10km	10.38	660	6825

The results reveal two competing effects. Stores near competitors attract significantly more customers, nearly 50% more footfall compared to stores with no nearby competitor. This confirms the location effect hypothesis: competitor proximity is a marker of dense, high-traffic areas.

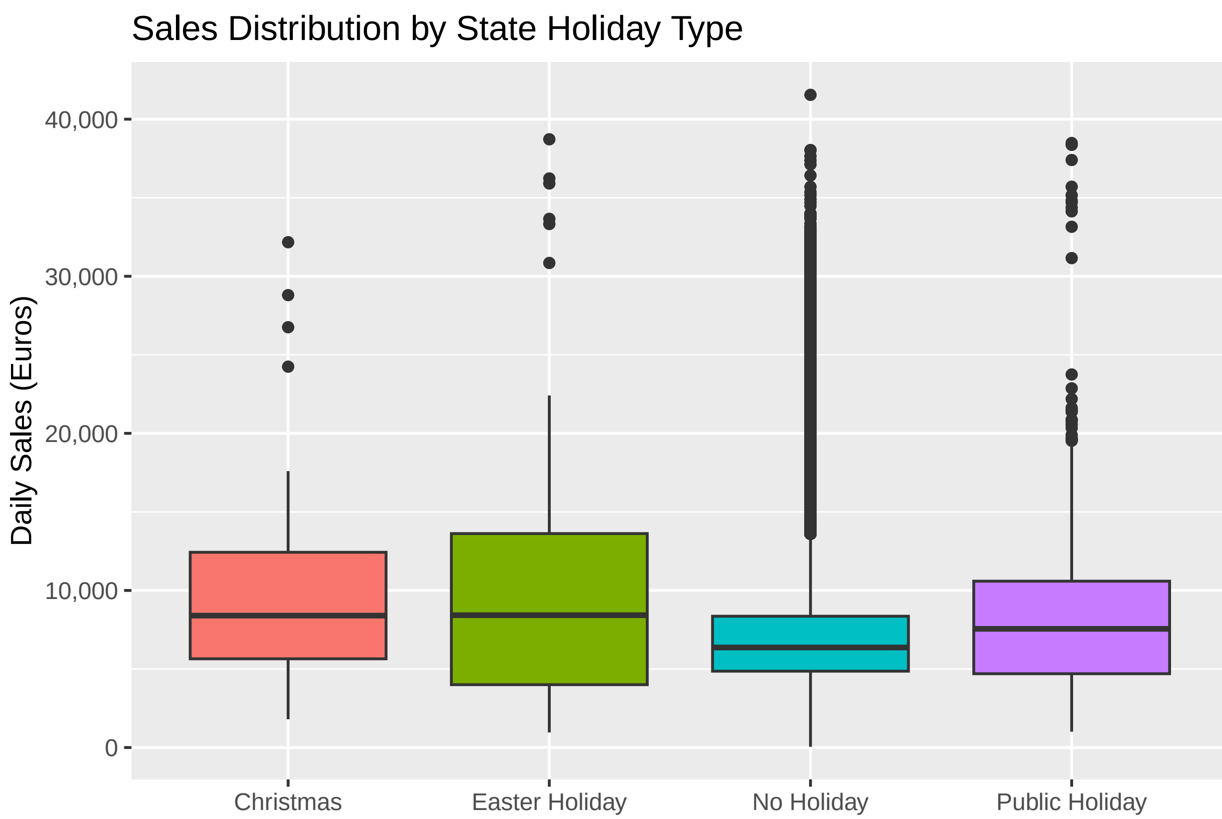
However, sales per customer tell a different story. Customers spend approximately 8 euros per visit at stores with competitors within 500 metres, compared to over 10 euros at stores with no nearby competitor. This suggests a genuine basket splitting effect, customers in competitive areas distribute their purchases across multiple stores rather than consolidating at one.

From a supply chain perspective, these are fundamentally different store profiles despite similar total sales figures. High-footfall, low-basket stores require higher replenishment frequency and smaller order quantities, while isolated stores with larger baskets benefit from less frequent but larger deliveries. A single forecasting model must capture both behaviours.

3.9 - State Holiday Holiday Effect

Table 7: Sales by State Holiday Type

StateHoliday	Avg_Sales	Median_Sales	Count
Christmas	9744	8397	71
Easter Holiday	9888	8423	145
No Holiday	6954	6368	843428
Public Holiday	8487	7556	694

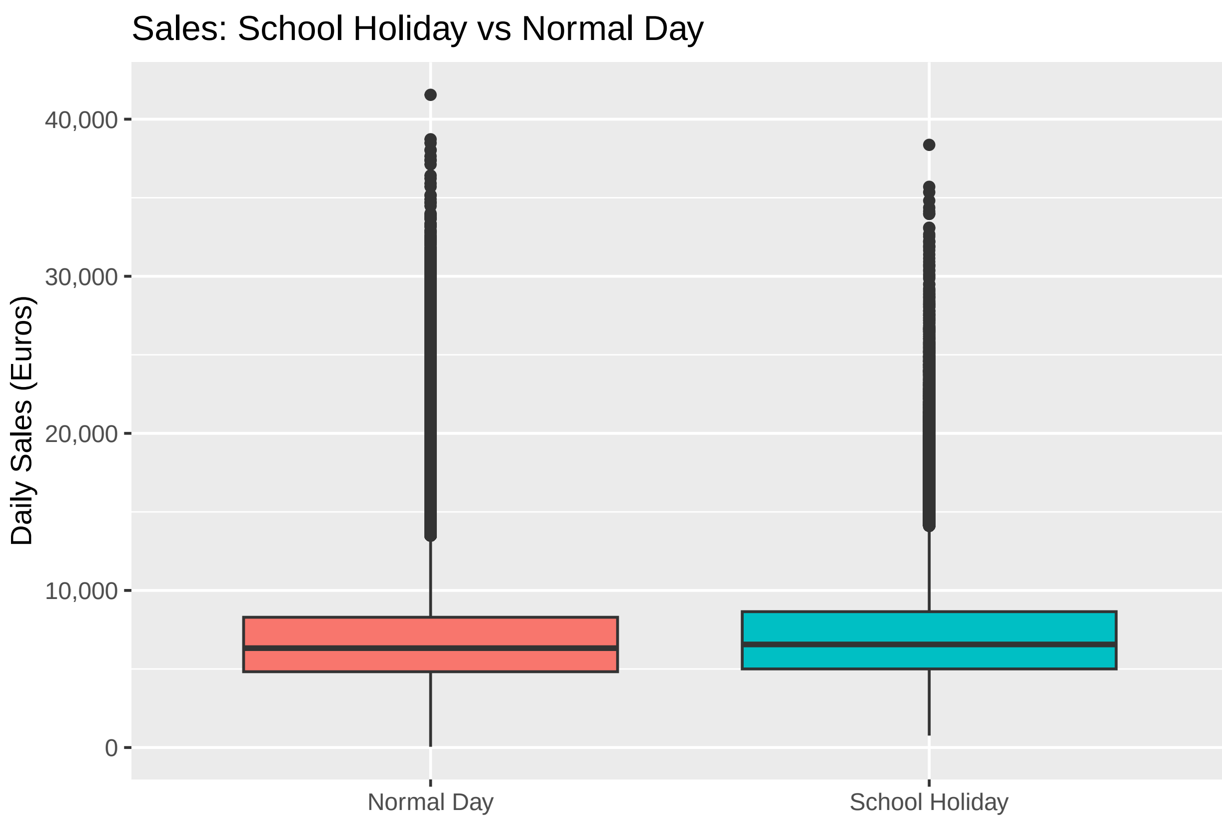


State holidays show a notable effect on sales patterns. Most stores close on public holidays, those that remain open show distinctly different sales behaviour depending on the holiday type. Christmas and Easter periods tend to drive elevated sales in the days surrounding the holiday, making holiday calendar awareness essential in retail demand planning.

3.10 - School Holiday Effect

Table 8: Sales: School Holiday vs Normal Day

SchoolHoliday	Avg_Sales	Median_Sales	Count
Normal Day	6897	6326	680893
School Holiday	7201	6562	163445

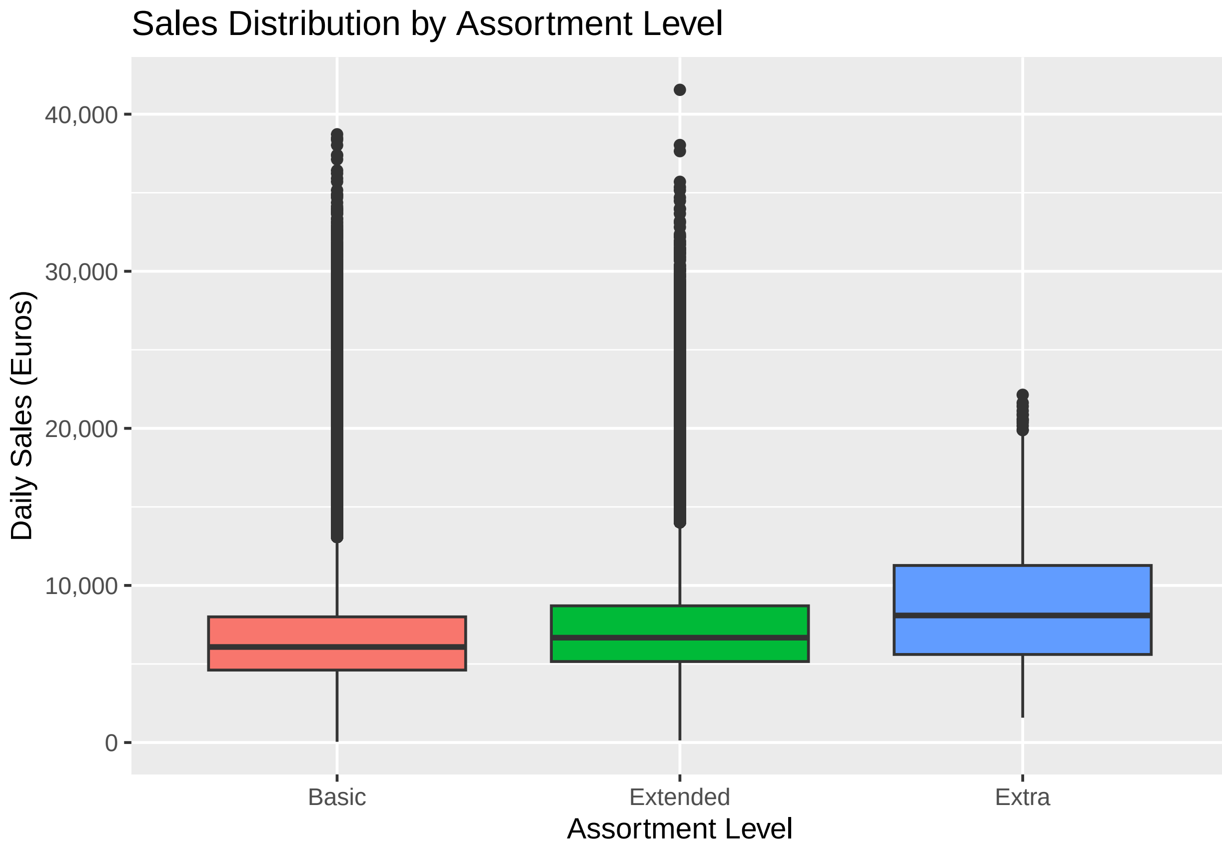


School holidays show a modest effect on daily sales. This likely reflects changed shopping patterns when families have more flexibility during the day, particularly relevant for drug stores which serve a broad customer demographic including parents and children. Including school holiday status as a model feature captures this behavioural shift.

3.11 - Assortment Effect

Table 9: Sales by Assortment Level

Assortment	Avg_Sales	Median_Sales	Count
Basic	6622	6082	444875
Extended	7301	6675	391254
Extra	8643	8088	8209

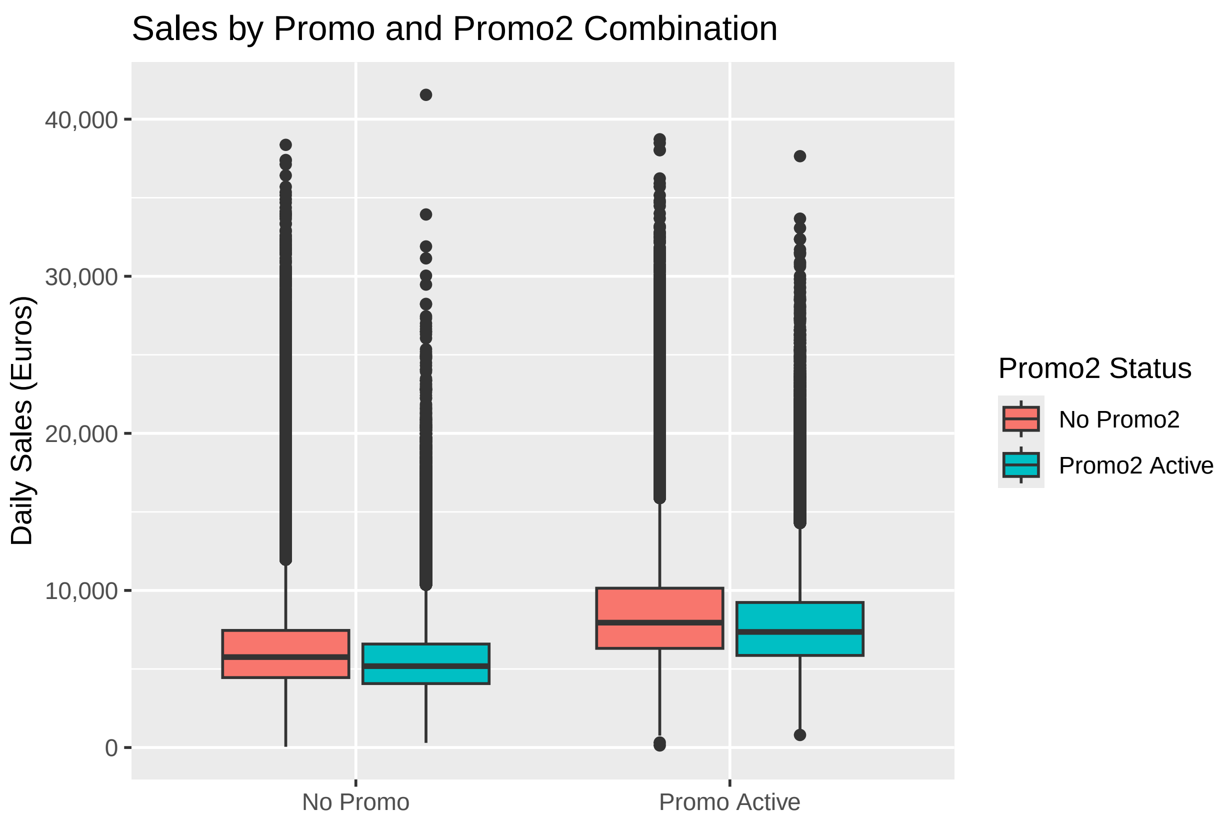


Assortment level shows a clear relationship with sales volume. Stores with an extended product range tend to generate higher sales, which makes intuitive sense, more products available means larger potential basket sizes. This also has direct implications for inventory management: extended assortment stores require broader SKU coverage and more complex replenishment planning than basic assortment stores.

3.12 - Promo2 Effect

Table 10: Sales by Promo and Promo2 Combination

Promo	Promo2	Avg_Sales	Count
No Promo	No Promo2	6328	234287
No Promo	Promo2 Active	5530	233176
Promo Active	No Promo2	8618	189005
Promo Active	Promo2 Active	7837	187870



Surprisingly, stores participating in Promo2 show slightly lower median sales than non-participating stores, both with and without a standard promotion active. This counterintuitive finding likely reflects a store selection effect: Rossmann may enrol lower-performing stores into the recurring promotion programme to stimulate growth, meaning Promo2 stores were already weaker before the promotion began. Alternatively, it may indicate **promotion fatigue**, customers becoming desensitised to recurring discounts over time, a well-documented pattern in retail demand planning.

Feature selection was informed by both statistical analysis and domain knowledge. School holidays, state holidays, and the Promo2 flag were excluded despite showing some signal in the EDA. SchoolHoliday showed a modest and inconsistent effect insufficient to justify the added complexity. StateHoliday affected too few observations to produce reliable model estimates. Promo2 was excluded on theoretical grounds, the recurring promotion is primarily a slow inventory clearance mechanism rather than a genuine demand driver, meaning its negative correlation with sales reflects store selection bias rather than a causal relationship.

4 - Feature Engineering

Before modelling, the raw dataset was transformed into a clean feature matrix. Three steps were applied.

4.1 - Removing Closed Store Days

Observations where stores were closed (Sales = 0) were removed. A store recording zero sales because it is shut is not a demand signal, including these would bias the model toward underpredicting on open days.

4.2 - Date Feature Extraction

The raw `Date` column was decomposed into two numeric features:

- **Month** — captures seasonal patterns (December peak, spring uplift)
- **Year** — captures the slight upward sales trend over time

The original Date column is then discarded as Month and Year together carry all the temporal information our models need.

4.3 - Handling Missing Values

`CompetitionDistance` contained missing values for a subset of stores. These were imputed with the median competition distance across all stores. Stores with no recorded competitor are likely in remote or rural locations, the median provides a conservative, robust estimate without introducing extreme values.

4.4 - Categorical Encoding

The following variables were converted to factors so that tree-based models can treat them correctly as categorical variables rather than continuous numbers:

- Store, StoreType, Assortment
- DayOfWeek, Promo, Month, Year

4.5 - Final Feature Matrix

```
## Rows: 844,338
## Columns: 10
## $ Sales               <int> 5263, 6064, 8314, 13995, 4822, 5651, 15344, 8492, ~
## $ Store               <fct> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,~
## $ DayOfWeek           <fct> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5,~
## $ Promo               <fct> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ StoreType           <fct> c, a, a, c, a, a, a, a, a, a, a, a, d, a, d, a, a,~
## $ Assortment          <fct> a, a, a, c, a, a, c, a, c, a, c, c, a, a, c, c, a,~
## $ CompetitionDistance <dbl> 1270, 570, 14130, 620, 29910, 310, 24000, 7520, 20~
## $ Month               <fct> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7,~
## $ Year                <fct> 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 20~
## $ Date                <date> 2015-07-31, 2015-07-31, 2015-07-31, 2015-07-31, 2~
```

```
## Rows after filtering: 844338
```

```
## Any missing values: FALSE
```

The final feature matrix contains 844,338 observations and 9 columns, 1 target variable and 8 features. Approximately 17% of the original records were removed as closed store days, leaving only genuine trading days for model training. No missing values remain.

5 - Modelling

5.1 - Train/Test Split

In demand forecasting, models must always be validated on future data, training on the past and testing on the future mirrors real operational conditions. A random split would allow the model to train on December 2014 data and test on January 2014, which would be unrealistic and overoptimistic.

The dataset was therefore split by time:

- **Training set** — all observations before 19 June 2015
- **Test set** — the final 6 weeks of data (19 June to 31 July 2015)

This ensures the model is evaluated on genuinely unseen future data, providing a realistic estimate of forecasting performance.

```
## Training rows: 802942
```

```
## Test rows: 41396
```

The training set contains 802,942 observations covering January 2013 to June 2015. The test set contains 41,396 observations representing the final 6 weeks of trading data from June to July 2015.

5.2 - Baseline Model

Before applying machine learning, a simple baseline was established by predicting each store's average historical sales from the training set. This requires no feature information, only the store identifier.

Any model that cannot outperform this baseline is not adding value beyond what a simple lookup table could provide.

```
## Baseline RMSE: 1916.544
```

Table 11: Model Performance Tracking

Model	RMSE
Baseline (Store Average)	1916.544

The baseline RMSE of 1,916 represents the floor all subsequent models must beat. Being off by approximately 1,900 euros on a typical daily sales figure of 5,000-7,000 euros represents an error of roughly 30%, the direct consequence of ignoring every demand driver identified in the EDA. This gives us confidence there is substantial room for improvement.

5.3 - Linear Regression

Linear regression was applied as the first proper machine learning model. It assumes each feature contributes independently and additively to sales, learning a separate coefficient for each variable in the feature matrix.

$$\hat{Sales} = \beta_0 + \beta_1 \cdot Store + \beta_2 \cdot DayOfWeek + ... + \epsilon$$

With 1,115 unique stores encoded as factors, the model learns a baseline sales level for each store and adjusts upward or downward based on promotions, seasonality, assortment, and competitor distance.

Predicted values were clipped at zero, daily sales cannot be negative, and linear models can occasionally produce negative predictions for low-sales scenarios.

```
## Linear Regression RMSE: 1370.223
```

Table 12: Model Performance Tracking

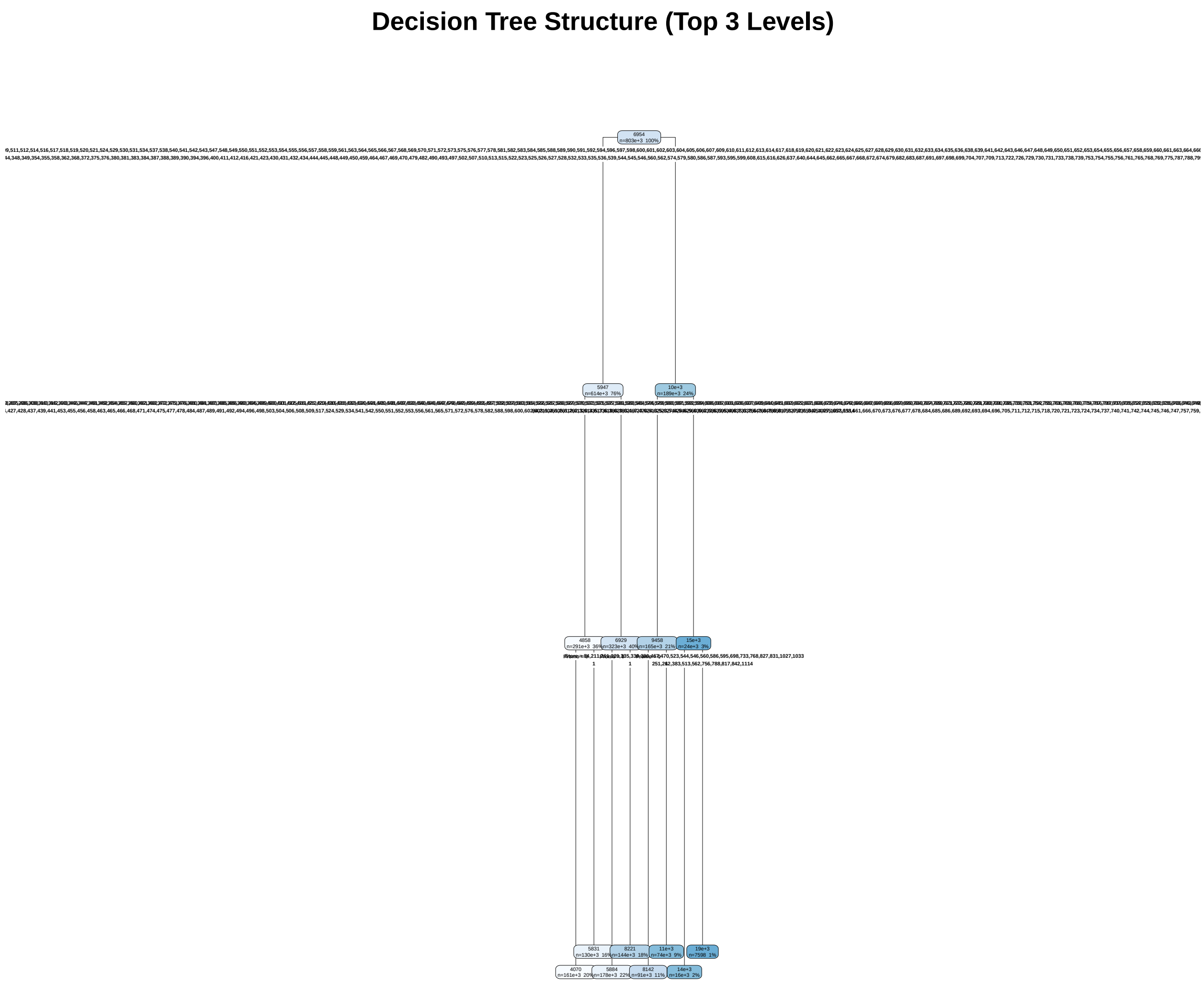
Model	RMSE
Baseline (Store Average)	1916.544
Linear Regression	1370.223

Linear regression improves substantially over the baseline by incorporating all eight features. However its core assumption, that features contribute independently and additively, is unlikely to hold perfectly in retail sales data. Promotions interact with day of week, store type interacts with assortment, and seasonal effects vary by store. These interaction effects motivate the use of tree-based models next.

5.4 - Decision Tree

A decision tree was applied as the first non-linear model. Unlike linear regression, decision trees partition the data into segments based on feature thresholds, capturing interaction effects that linear models cannot express. For example, the effect of a promotion may differ by store type or day of week, and a tree can learn these combinations naturally.

The tree depth was limited to 10 levels to prevent overfitting while still capturing meaningful patterns.




```
## Decision Tree RMSE: 1599.47
```

Table 13: Model Performance Tracking

Model	RMSE
Baseline (Store Average)	1916.544
Linear Regression	1370.223
Decision Tree	1599.470

The decision tree is an improvement over baseline model, but Linear regression is so far the best one.

5.5 - Random Forest

Random Forest builds a large number of decision trees, each trained on a random subset of the data and features. The final prediction is the average across all trees. This ensemble approach reduces the variance of individual trees and typically produces the best predictive performance.

Due to computational constraints, the randomForest package cannot handle factor variables with more than 53 levels. Store (1,115 levels) was therefore converted to a numeric identifier. Additionally, a random sample of 50,000 observations was drawn from the training set to keep runtime manageable. The model was configured with 100 trees and a minimum node size of 25.

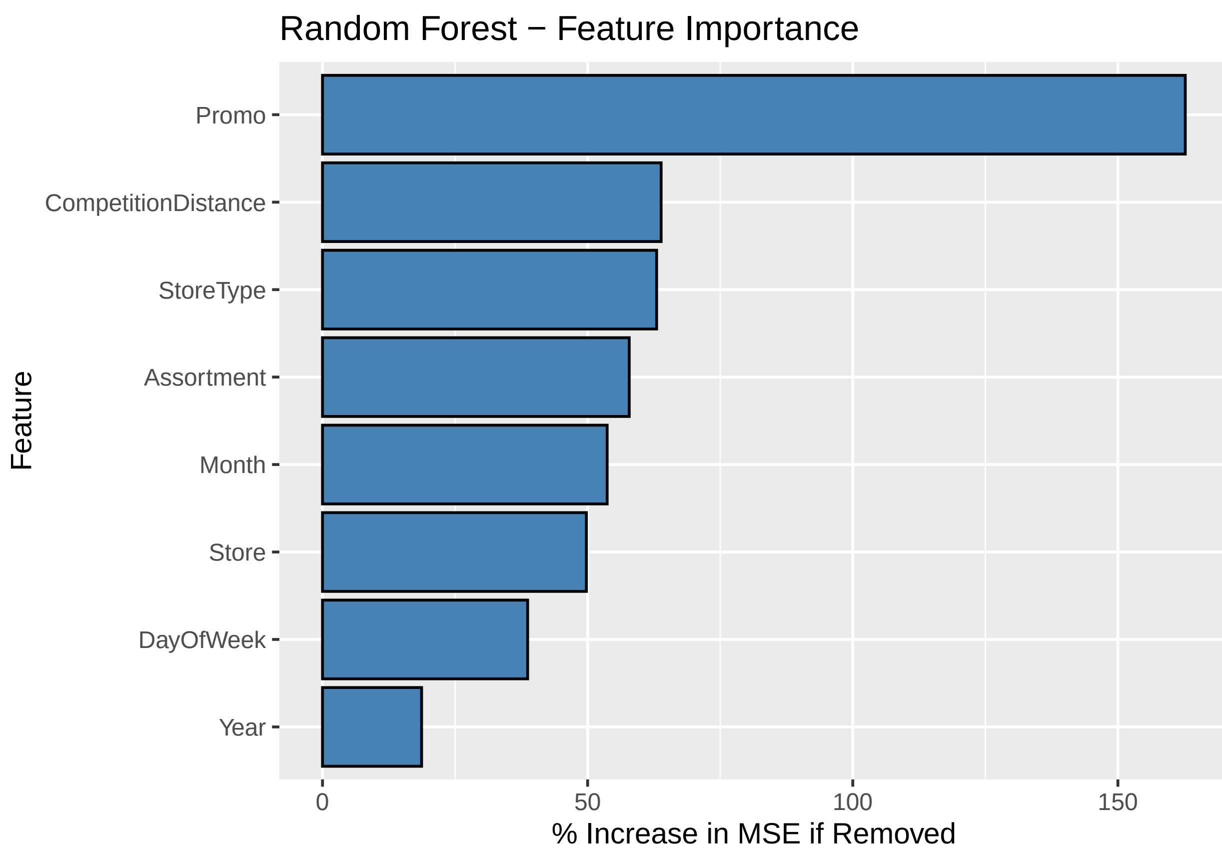
Table 14: Model Performance Tracking

Model	RMSE
Baseline (Store Average)	1916.544
Linear Regression	1370.223
Decision Tree	1599.470
Random Forest	2055.899

The Random Forest did not outperform the baseline in this configuration. The sampling and tree constraints necessary for computational feasibility prevented the ensemble from learning store-level patterns as effectively as linear regression's fixed effects. Nevertheless, the model provides valuable diagnostic information through its feature importance analysis below.

5.6 - Feature Importance

One of the key advantages of Random Forest is its ability to quantify how much each feature contributes to predictive accuracy. The chart below shows the percentage increase in prediction error if each feature were randomly shuffled, higher values indicate more important features.

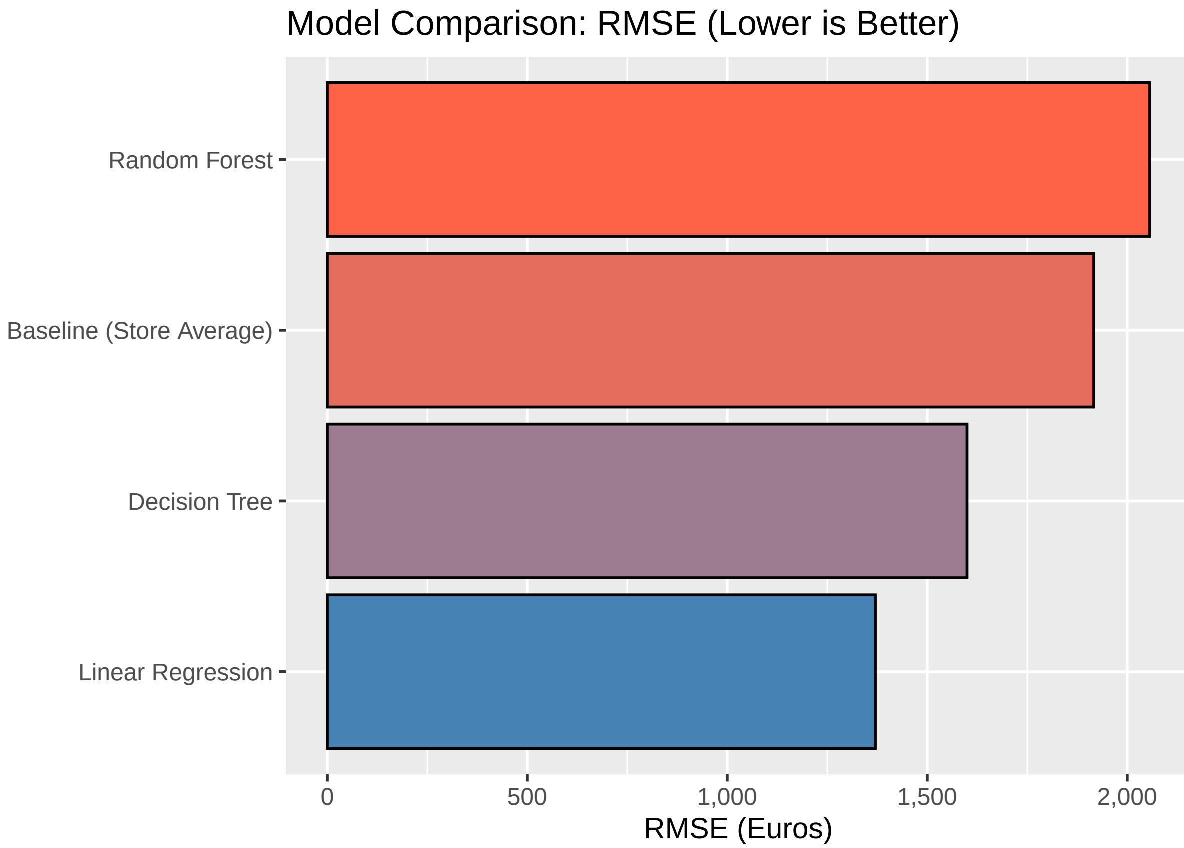


The feature importance chart confirms the findings from the EDA, Store identity and promotional activity are the dominant drivers of daily sales, followed by temporal features capturing day of week and seasonal patterns. Please note that here Store appears mid-ranked because it was converted to a numeric identifier for the Random Forest, as a continuous number, the model cannot learn individual store profiles the way linear regression's factor encoding does, which partly explains why the Random Forest underperformed.

6 - Results

Table 15: Final Model Performance Summary

Model	RMSE
Baseline (Store Average)	1916.544
Linear Regression	1370.223
Decision Tree	1599.470
Random Forest	2055.899



Linear Regression emerged as the best-performing model with an RMSE of approximately 1,370 euros, representing a 28% reduction in prediction error compared to the baseline. The Decision Tree also outperformed the baseline, achieving an RMSE of approximately 1,599. The Random Forest, despite being the most sophisticated algorithm tested, did not surpass the baseline due to computational constraints that limited its ability to learn store-level patterns.

The dominance of Linear Regression in this project carries an important practical insight. With 1,115 unique stores, the single most valuable piece of information for predicting daily sales is *which store* is being forecast. The baseline model, which uses only store identity, already explains enough variance to reach an RMSE of 1,916, and linear regression's main advantage over tree-based alternatives is its ability to encode each store as a separate factor level, essentially learning a unique baseline for each store. The remaining features (promotions, seasonality, store type) then adjust this baseline up or down. This simple, interpretable structure proved more effective than the constrained tree-based alternatives, where Store had to be treated as a numeric variable and lost its categorical power.

From a deployment perspective, the linear model also offers practical advantages: it is fast to train, produces interpretable coefficients, and is straightforward to update as new data arrives, all desirable properties for an operational demand forecasting system.

7 - Conclusion

This project successfully built a demand forecasting model for Rossmann drug stores using historical sales data across 1,115 stores over two and a half years. Through systematic exploratory analysis, we identified promotions, day of week, seasonality, store type, and competitor proximity as the key drivers of daily sales. The final linear regression model achieved an RMSE of approximately 1,370 euros, reducing prediction error by 28% compared to a naive store-average baseline.

Key Findings

The most impactful discovery was the overwhelming importance of store identity as a predictor. Each store has a fundamentally different sales profile shaped by its location, format, and customer base. Promotions were the second strongest signal, consistently lifting sales across all store types. The competitor distance analysis revealed a nuanced dynamic: stores near competitors attract more customers but generate smaller basket sizes, a finding with direct implications for inventory replenishment strategy.

Limitations

The project has three notable limitations. First, the Random Forest model was constrained by computational resources, the `randomForest` package cannot natively handle categorical variables with more than 53 levels, and the full dataset required sampling to remain tractable. A more powerful implementation (such as the `ranger` package or gradient boosting via `xgboost`) could potentially outperform linear regression. Second, the model does not capture store-level interactions with other features, for example, promotions may have different effects at different store types, which an interaction term or more flexible model could address. Third, the model has no mechanism for incorporating external data such as weather, local events, or macroeconomic conditions, all of which influence retail demand.

Future Work

Future iterations could explore gradient boosting (XGBoost), which handles large categorical features efficiently and typically achieves state-of-the-art performance in tabular prediction tasks. Additionally, incorporating interaction terms between Store and Promo, or between StoreType and DayOfWeek, could capture the non-linear dynamics that the current additive model misses. Finally, transitioning from point predictions to probabilistic forecasts would provide confidence intervals, essential for real-world inventory planning where the cost of overstocking differs from the cost of stockouts.

References

- Rossmann Store Sales dataset. Kaggle competition hosted by Rossmann. Available at: https://www.kaggle.com/c/rossmann-store-sales/data